Question 1

a) Restriction fragment length polymorphism, RFLP, refers to differences in fragment length generated in different individuals by restriction enzymes.

Figure 1.1 below shows an autosomal dominant disease segregating in a family. Genomic DNA from each family member was assayed for RFLP for *Eco*RI sites, on Southern blots using probes derived from the gene.

The results for all individuals, except those for the foetus, are shown below the pedigree chart. The sizes of the DNA fragments obtained are as indicated.

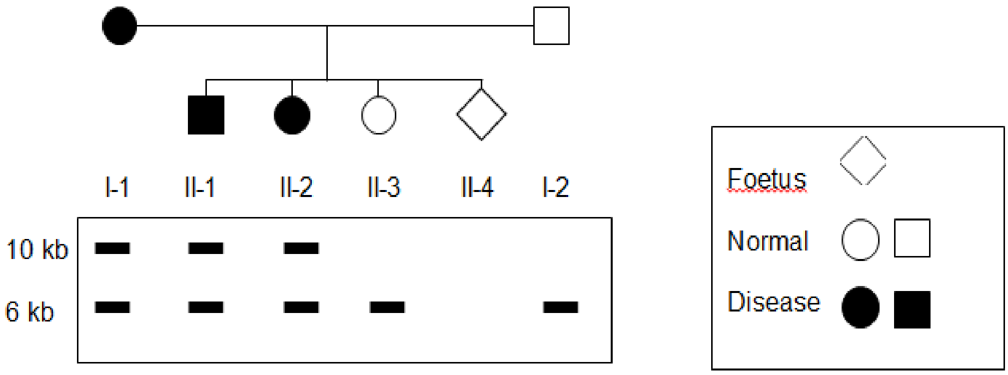
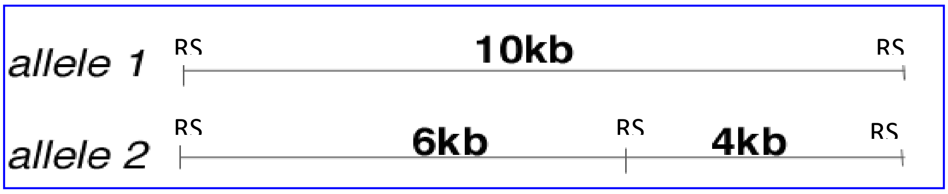


Figure 1.1

i) The diagram below shows the 2 possible alleles of the gene. Indicate on the diagram the possible location recognized by the probe that was added. [2]



RS= restriction site

Which RFLP fragment did the mother (I-1) contribute to the diseased children? [1]

10 kb fragment

ii) What is the probability of the foetus having the disease? Explain your answer. [2]

- 50% or ½ probability
- Mother's genotype is heterozygote; father's genotype is homozygous recessive so chance of receiving dominant allele ½

b) A different study was carried out on another family. **Figure 1.2** below shows a Southern blot with an RFLP detected by using a probe for a sequence located on a sex chromosome. The variant exhibits two forms, one at 8 kb and one at 6 kb. The mother, father and daughter are normal.

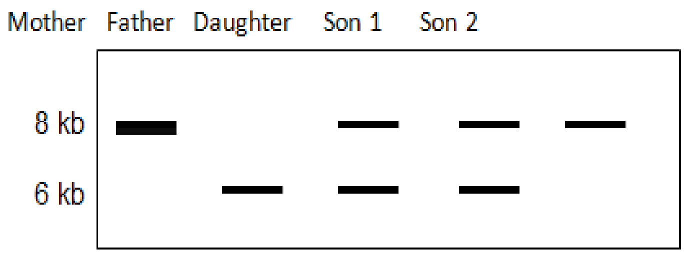


Figure 1.2

From **Figure 1.2**, deduce which son is abnormal and explain how you arrive at our answer. [3]

1. Son 1

2. Sex linked disease. Y chromosome does not carry any allele FOR SEX LINKED diseases. Hence male should only exhibit one band.

3. Son 1 is a case of XXY/ Klinefelter's syndrome

c) RFLP is widely used as a tool for disease detection. In disease detection, genetic markers are often used to predict if a person is likely to suffer from a particular disease.

i) Explain the features of genetic markers that allowed them to be used for the indirect analysing of the inheritance of disease. [3]

1 known sequence (DNA/gene); with known location on a chromosome;

2 genetic marker found close to the gene of interest;;

3 3 little probability of crossing over
hence presence of genetic marker = presence of disease

d) DNA fingerprinting through the use of RFLP is a technique employed by forensic scientists to assist in the identification of individuals by their respective DNA profiles.

i) Explain why genes especially protein-encoding genes are seldom used for genetic fingerprinting. [2]

1. sequence of protein-encoding genes are conserved; due to its important biological function;

2. not able to differentiate between individuals if you use protein encoding region

[Total: 11 marks]

Question 2

Flavr Savr is a genetically modified tomato. It is made more rot-resistant by adding an antisense gene to interfere with the production of the enzyme polygalacturonase (PG). This enzyme causes the softening of the fruit by degrading pectin.

PG production during fruit ripening was measured in plants with one and with two antisense genes, and in normal plants. The results are shown in **Figure. 2.2.**

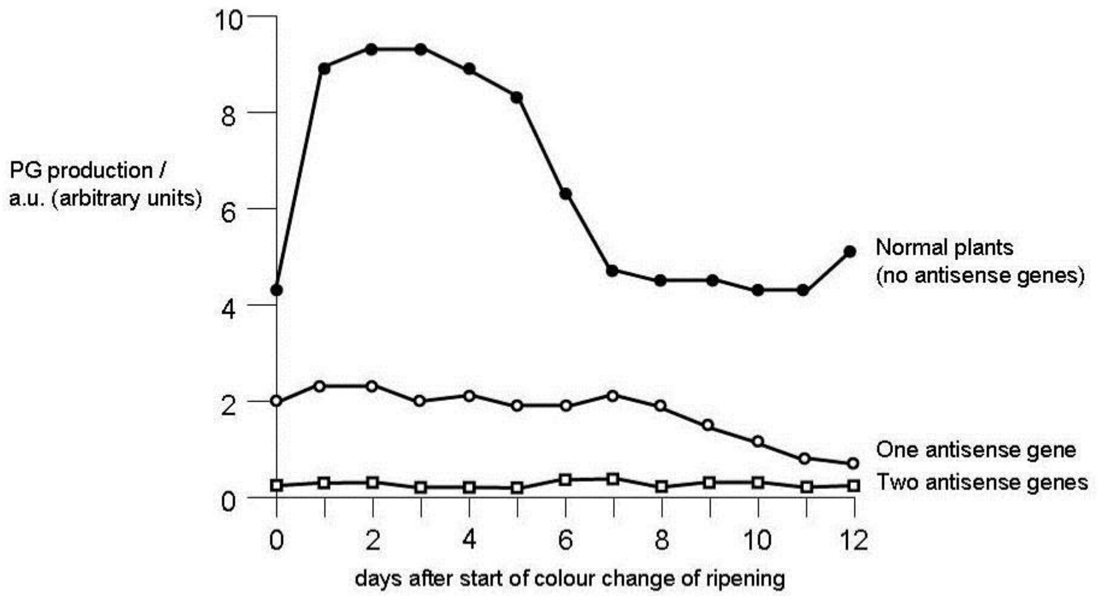


Figure 2.2

With reference to **Figure 2.2**, describe and explain the difference between normal tomato plants and transformed plant with one anti-sense gene. [4]

- PG production in normal plants higher than plants with one antisense gene, min ~4 a.u./max~9 a.u. as compared to about 2 a.u.;
- antisense gene is being expressed in transformed plant cells, resulting in formation of duplex RNA;
- less translation initiation complex formation / decreased binding of ribosome to mRNA;
- resulting less translation of mRNA into PG proteins;

Question 3

a) Cystic Fibrosis (CF) is an autosomal recessive genetic disorder that affects predominantly the lungs and other mucosal organs in the body. A technique that measures the nasal potential difference has been investigated recently as a possible diagnostic tool. When chloride ion channels open, a potential difference will be recorded as a positive spike in the potential difference against time graph.

In the lab, researchers performed experiments on two groups of nasal epithelial cells, X and Y. To observe changes in nasal potential difference, the researchers perfused a solution of amiloride (Am) into the nasal activity, followed by the subsequent perfusion of the nasal cavity with a low chloride solution (LC).

Amiloride serves to block sodium absorption, resulting in a reduction of the potential difference. This provides an electrical gradient for the secretion of chloride, which is stimulated by the biochemical gradient provided by a low chloride solution.

The potential difference against time graph was plotted as shown in Figure 3.1 below.

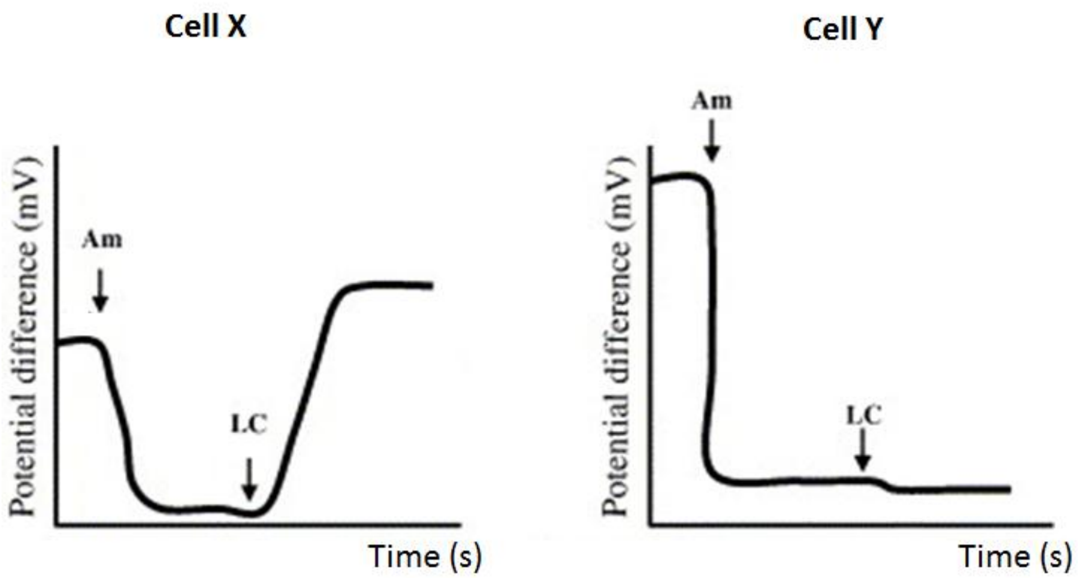


Figure 3.1

i) Explain what is meant by “autosomal recessive” genetic disorder. [2]

- Condition arises only when both alleles of the gene are mutated/ Only expressed in homozygotes
- Gene is carried on chromosomes other than sex chromosomes.

ii) With reference to Figure 3.1, identify the cell with cystic fibrosis. Explain your answer. [3]

- Cell Y
- The graph for cell Y did not show a positive spike/ change in potential difference unlike the graph for cell X when LC (low chloride solution) was perfused into nasal cavity, hence showing that chloride channels failed to open properly and chloride ions could not exit cell.
- Due to mutation of the CFTR gene that encodes a mutant CFTR channel protein that does not allow movement of chloride ions thus no change in potential difference.

iii) Describe how cystic fibrosis can be cured by gene therapy using a non-viral vector. [4]

- In vivo gene therapy method
- Functional allele coding for CFTR on chromosome 7 is isolated from a normal individual
- Aerosol spray containing liposomes with the normal allele can be sprayed into the nose and moith of CF patients
- Membrane of liposomes fuse with the cell membrane of target lung epithelial cells, releasing the normal CFTR allele into cytoplasm via endocytosis
- Normal CFTR protein is expressed and incorporated into target cell's membrane.

iv) Suggest two technical challenges preventing the effectiveness of gene therapy for cystic fibrosis. [2]

- Gene therapy is short-lived as target cells for CF treatment are epithelial cells that are constantly being shed, thus repeat therapy is likely to be required.
- Use of viral vectors (adenovirus) can trigger host immune response.
- Other less accessible organs like the liver and pancreas are also affected by CF and is difficult for therapy to reach effectively.
- Liposomes are not specific in their action and thus have a low transfection efficiency.
  (max 2)

b) The procedure of current gene therapy for severe combined immunodeficiency (SCID) is illustrated in Figure 3.2 below.

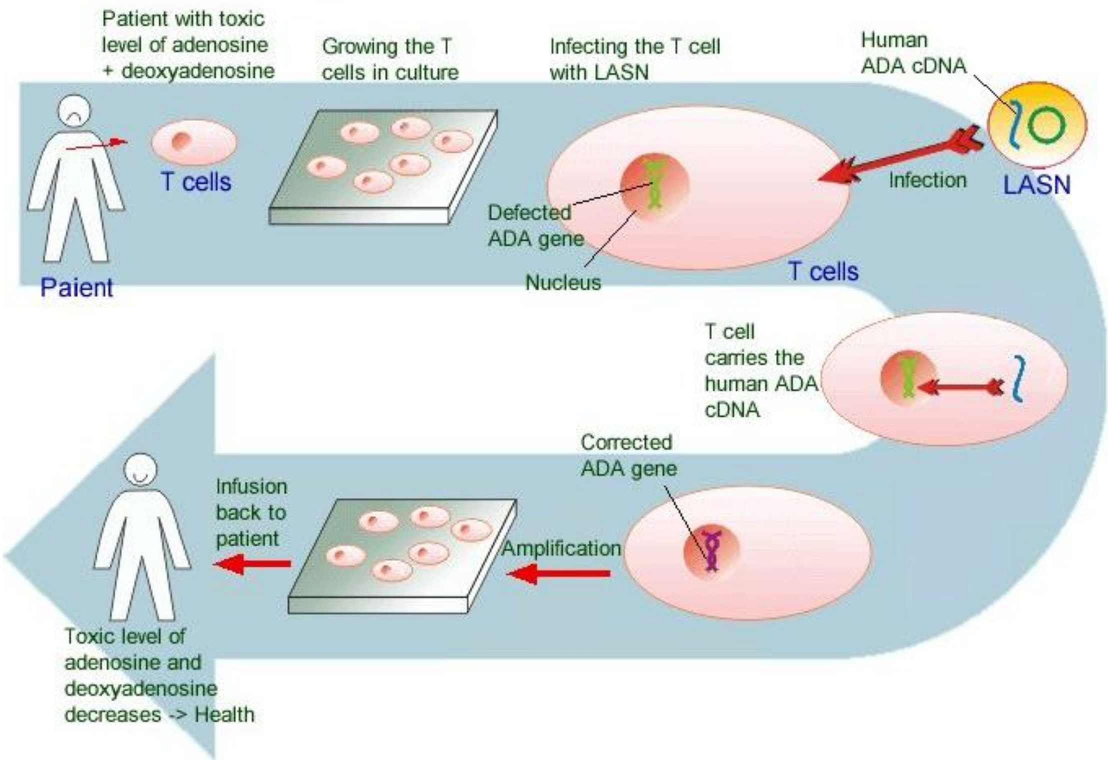


Figure 3.2

i) State two advantages of using the ex vivo method for gene therapy that is illustrated in Figure 3.2. [2]

- Allows for selection of successfully transfected cells;
- Ease of transfection and culture of T lymphocytes in the lab;
- Specific approach ensures that retrovirus targets the T lymphocytes instead of other cells in the body.

ii) Suggest and explain how you can improve the above gene therapy procedure in order to sustain long-term success of the treatment. [2]

- Use blood stem cells extracted from bone marrow apart instead of T lymphocytes.
- Blood stem cells are able to differentiate into T lymphocytes and show long-term self renewal so as to provide a sustained cure

Question 4

a) There has been rampant speculation that the consumption of genetically modified (GM) crop and livestock has impacted birth rates in the USA negatively. A study was carried out in the USA to investigate the claim. Figure 4.1 shows a graphical plot of the data obtained.

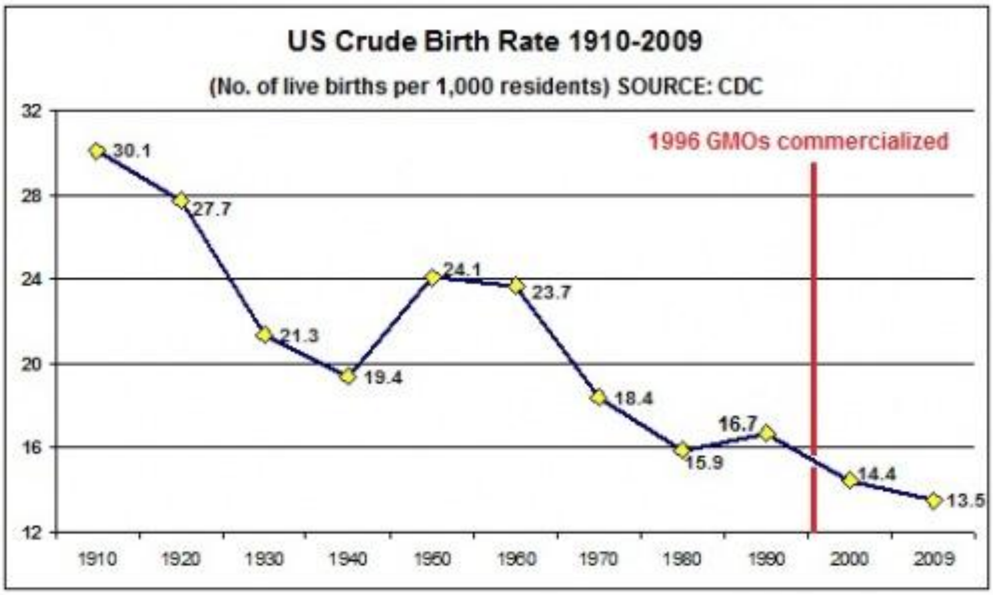


Figure 4.1

i) With reference to Figure 4.1, comment whether the speculation can be justified by the data. [3]

- Speculation cannot be justified
- From 1910 to 2009, the US crude birth rate has been on a decline from 30.1 to 13.5 with only two exceptions between 1940 to 1950 and 1980 to 1990.
- Also, commercialisation of GMOs in 1996 did not lead to a significant decline in birth rate ten years later in 2006 (approximately 15.0 to 13.0), as compared to the sharpest decline in birth rate from 1920 (27.7) to 1930 (21.3), thus shows that other factors could have been responsible for the declining birth rate.

ii) Describe two threats that GM food can pose to human health. [2]

- Vectors used in the process of engineering GM food might contain antibiotic resistance that may be passed on to bacteria like E Coli in the human gut.
- GM food might contain allergens and can cause unwanted allergic reactions in the body.

b) The effectiveness of CpTi against insect pests was tested using two groups of tobacco plants.

- Plants genetically engineered to express the allele coding for CpTi.
- Control plants which had been genetically engineered by inserting the allele coding for CpTi in the wrong orientation. These plants did not produce CpTi.

Equal numbers of each type of plant were infested with equal numbers of newly hatched larvae of an insect pest and left for seven days.

The mean percentage of leaf area eaten and the mean mass of surviving insects were then found. The results are shown in Figure. 4.2.

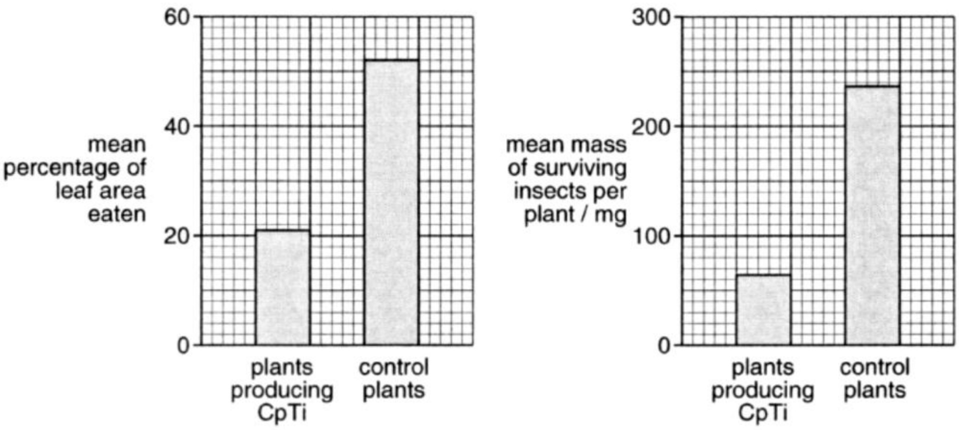


Figure 4.2

With reference to Figure 4.2,

(i) Describe the effect of CpTi on yield of tobacco leaves. [2]

- Has a smailler mean percentage of leaf area eaten
- 52% in control, 21% in plants producing CpTi

(ii) Suggest an explanation of the effect of the trypsin inhibitor, CpTi, on insects.

- Refer to the lack of digestion of protein.
- Enzyme inhibition, hence active sites not able to bind to substrate.
- Insects would dies of malnutrition or starvation

[Total: 10 marks]

Section B (20 marks)

Free-response question

Write your answer on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

a) Outline the large scale production of a **named** important protein by genetic engineering. [10]

	1. Construction of recombinant DNA molecule (foreign + vector DNA) • **Source of gene - mRNA for insulin [2 marks max]** • **Isolate mRNA from beta cells of pancreas;** • **RT is used to reverse transcribed mRNA to form complementary DNA (cDNA);** • **DNA pol then synthesizes the second DNA strand;** • **Mentioned use of adaptors/ linkers to create sticky ends (since cDNA will be blunt);** **b. Creation of recombinant DNA molecule [2 marks max]** • **Vector and insulin gene must be cleaved with same restriction enzyme at specific recognition sequence/site;** • **So that complementary base pairing can form via hydrogen bonds between insulin gene fragment and vector;** • **Example (of vector/ restriction enzyme);** • **DNA ligase is added to seal phosphodiester bond;** **2. Introduction of recombinant DNA molecule into living host (e.g. bacteria) [2 marks max]** • **Via transformation;** • **Method of transformation;** • **Reason of method (to create permeability/ transient enlargement of pores);** • **Purpose of living host (provide enzyme machinery);** **3. Multiplication of recombinant DNA molecule and cell growth/ division; [1 mark]** **4. Screening/ Selection & identification of clones with recombinant DNA molecules [2 marks max]** **Insertional inactivation;** **Lac Z/ Blue-white selection or antibiotic selection;** **Description of either lac Z or antibiotic selection;**

	5. **Purification of insulin product [2 marks max]** **Upscale/ production/ fermentor/ bioreactor (max 1 mark);** **Post-translational modification/ cleave fusion polypeptide with CNBr / combining the 2 chains/polypeptide via disulphide bonds; [max 1 mark]**

b) Using specific example(s), compare viral and non-viral delivery systems in the delivery of a functional CFTR allele.[6]

Introductory statement:

- **Viral delivery systems for treatment of cystic fibrosis involves the use of adenoviruses and adeno-associated viruses; Non-viral delivery system for treatment of cystic fibrosis involves use of liposomes OR gene gun;**

Similiarities:

- **Both adeno-associated viruses and liposomes do not produce immune response in the body. [1]**
- **Both adenoviruses and non-viral do not result in long term expression of the gene and these methods does not result in integration of normal gene into host cell genome. [1]**

Differences:

Basis of comparison	Viral methods	non-viral methods	
effectiveness of gene transfer	• adenoviruses naturally infect lungs, and hence are effective for delivering the gene	• both liposome and gene gun are less effective	[1]
long term expression of gene	• use of adeno-associated virus can result in integration of normal gene into host cell genome, and may result in long term expression of gene.	• low	[1]
immune response	• use of adenoviruses can cause body to produce immune response	• minimal host immune response is generated	[1]
side effects	• not known	• minimal	[1]

c) Describe the advantages and limitations of the Polymerase Chain Reaction (PCR).
[4]

Advantage:

- **sensitive;**
- amplification possible from **trace amount of template DNA;**
- fast/efficient method of testing as 35 – 40 PCR cycles can be completed in several hours/(ref to short times required for each PCR segment); [R: “fast/efficient” without any elaboration];
- use of **thermostable** DNA polymerase allows for automation of procedure without need to add polymerase for every cycle;
- allows **amplification/replication** of selected target region of DNA flanked by/usingspecific PCR primers;
- **Programmable thermal cycler** allows automation of PCR;
- faster method of amplifying/cloning DNA sequence **than conventional cell-based cloning;**

Reject: create many copies of DNA using small sample, sensitive allowing amplification of small DNA

Disadvantage:

- sensitive to contamination of template DNA by **foreign/non-target nucleic acids**;
- may give false positives;
- requires careful handling to prevent contamination;
- by non-target nucleic acids;
- Use of **<u>Taq DNA polymerase;</u>** may result in misincorporation errors in PCR products; as (Taq DNA polymerase) lacks proof-reading activity / 3’□5’ exonuclease activity (to correct misincorporated bases);
- Needs preliminary knowledge of part of/nature of target sequence; before specific primers can be designed;
- Mutations in regions complementary to primer(s) may cause reduction in annea**ling/amplification efficiency;**
- PCR products labile/unstable as they are not associated with histones;

H2 P3 Planning Question - Mark Scheme Summary

Theoretical basis of experiment [max 3]:

1. Differences in DNA sequences among different individuals →differences in the number of restriction site locations
2. Basis of RFLP: SAME Restriction enzyme digest → different number and length of restriction fragments
3. If Bernard Soon is indeed the child of the billionaire, then his RFLP bands should match either the billionaire's or his mother's/his bands should eb accounted for by the billionaire or his mother..

Procedure:

Extraction of DNA [max1]

1. Break the tissues: pestle and mortar.
2. Lyse cells: DNA extraction buffer.
3. Centrifuge, add RNase and protein precipitation buffer → obtain pure DNA

Polymerase Chain Reaction [max 1]

4. Taq polymerase.
5. Heat DNA mixture (95 ^{o}C)
   → separate double-stranded DNA into 2 complementary single-stranded DNA.
   → break hydrogen .
6. Cool (55 ^{o}C)
   → primer hybridization to DNA template, primer provide 3'OH end
7. (72 ^{o}C) Replication and extension of the DNA in 5' to 3' direction
8. Each cycle is repeated. →exponential increase of target DNA sequences.

Restriction Enzyme Digestion [max 1]

9. Use same restriction enzyme → yield restriction fragments.

Gel Electrophoresis [max 1]

10. Negatively charged DNA molecules will migrate toward the positive end of the field (anode).
11. Separated according to their molecular size.
12. The smaller molecules → move faster than the larger one

Southern Blot [max 1]

13. Transfer DNA bands to a piece of nitrocellulose membrane by capillary action
14. The nitrocellulose membrane: replica of banding pattern

Probing [max 1]

15. Denature DNA into single-strands using NaOH
→ break hydrogen bonds between the complementary base pairs of double helix
16. Add single stranded radioactively labelled DNA probe complementary to gene of interest..

Visualization [max1]

17. The hybridized probe is detected by autoradiography.
18. Exposure to X-ray →image of radioactively labelled DNA bands.

Comparing DNA bands [max 1]

19. Compare DNA banding patterns

Conclusion [max1]

20. If Bernard Soon is indeed the child of the billionaire, then his RFLP bands should match either the billionaire's or his mother's/his bands should eb accounted for by the billionaire or his mother.